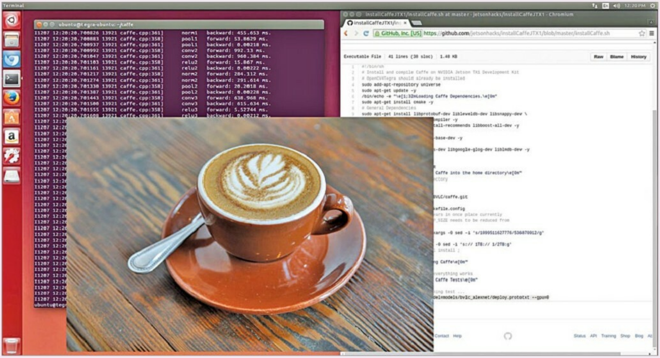

Fig. 2: Caffe deep learning framework

Various layer types

- Image data layer for reading raw images
- Database layer for reading data from LevelDB or lightning memory mapped database
- HDF5 input layer for reading HDF5 data
- HDF5 output layer for writing data as HDF5
- Window data layer for reading window data file
- Memory data layer for reading data directly from memory
- Dummy data layer for static data and debugging

the spatial input as a vector with a specific dimension.

Getting down to the basics of video capturing, layers such as convolution get into action and are responsible for producing one feature map in the output image. Convolution is an important operation in signal and image processing, and in Caffe, pooling, cropping, edge detection and transformation processes take place in a sequence, followed by deconvolution and loss of learning.

Caffe learns about the real world

Software like Caffe learn about the real world by specifying a certain level of badness, driven by a factor called loss function. A loss function specifies the goal of learning by mapping parameter settings to a scalar value. Now, the goal of learning in machines is to find a setting of weights that minimise the loss function.

In Caffe, loss is computed by the forward pass of the network where each layer takes a set of input (bottom) blobs and produces a set of output (top) blobs. In simple words, the model works by comparing the loss while generating the gradient and then incorporating the gradient into a weight update that tries to minimise the loss. This is done by a solver in Caffe.

Significance of deep learning

Machines have made our lives easier and now we are one step closer to making these think. With deep learning, deep networks and tools like Caffe, it is now possible to make machines learn the way we humans have learnt in practical applications of speech recognition and image classification, among others.

In reality, we are building features into machines using such specialised software and letting these connect to smart, artificial neural networks and frameworks to learn for themselves.

Significantly, these machines not only learn the features but run based on real-world events, where Caffe could work on cameras for combining, classifying and, to an extent, making intelligent decisions like informing concerned authorities through email or message alerts in places like a bank ATM, in case of unrecognised movement or threat.

With improvements in processing power and graphics, the bottlenecks of computation are being overcome over the years. Deep learning has already branched out to many other use cases. Text analytics, time-series analytics, image processing and real-time threat detection using video or motion detection are areas that are using deep learning techniques. EFY

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